

Turbulent nanofluid flow through a solar collector influenced by multi-channel twisted tape considering entropy generation

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Abstract. To enhance the performance of solar collectors, multi-channel twisted tape has been employed in this paper. Our main focus has been on the entropy generation of nanofluids. Inserting such device separates the nanofluid and a longer duration in pipe occurs. The number of channels and revolutions, the Reynolds number and the pipe diameters were considered as effective parameters. Results pointed out that more turbulence mixing occurs with greater values of the diameter ratio. Thermal irreversibility augments when reducing the number of channels. Because of more nanofluid gliding near the surface, frictional irreversibility enhances with increasing pumping power.

1 Introduction

The visual characteristics of nanoparticles went through an immense transformation owing to the tiny amount of nanoparticles effects, quantum effects, special massive surface area effects and the haphazard style of atomic classification. In order to reinforce conduction naturally, they added nanoparticles to the base liquids, which were very well received [1]. The analysis of slip impact on the transient nanoparticle flow including solar radiation was done by Kundu and Sarkar [2]. The end result of that analysis pointed out that while the flow gradually gets low, the expansion of surface slip is observed and it enhances the temperature profile. Anyhow, thermal slip manages to reduce the temperature yet increasing Nu . With thermal radiation in existence, Butt and Ali [3] studied the effect of hydro magnetic convection movement via a permeable plate. By utilizing nanofluids in the solar collector, Sheikh *et al.* [4] performed a relative research of the Atangana-Baleanu fractional derivative (ABFD) and Caputo-Fabrizio fractional derivative (CFFD). The effects of solar heating on fluid behavior surrounding a circular cylinder in porous medium were investigated by Mabood *et al.* [5].

An exploration was done on the behavior of narrow film over an extended plate by Khan *et al.* [6] and they also studied the effects of thermophoresis and thermal radiation. Sheikholeslami *et al.* [7] analyzed the phase change process in solar heat exchangers. To get more efficiency, they utilized fins and nanoparticles. Sundar *et al.* [8] carried out a ground-breaking experiment with Al_2O_3 nanofluid flow within solar collectors that had inserts of twisted tape. The results of the experiment demonstrate that when the rate of mass flow is at 0.083 kg/s, they see a 58% enhancement of effectiveness with no inserts but with inserts that have the highest ratio the percentage on enhancement is 76. Helical twisted tape has been imposed in a pipe by Sheikholeslami *et al.* [9]. They analyzed turbulent flow with the RANS model. While taking into consideration the effects of absorption and the heat generation, Hussein *et al.* [10] have done an examination on a 3-dimensional convective flow of nanoparticles along an extending plate with porous media. The CNT movement along a sheet was pointed out by Soomro *et al.* [11]. They exhibited entropy generation in the presence of the MHD. The exergy treatment of a magnetic nanofluid within a porous tank was illustrated by Sheikholeslami [12]. His outputs were obtained via CVFEM. With the aid of a stretching cylinder, Hayat *et al.* [13] pointed out the behavior of a melting parameter in a stagnation-point flow of a carbon nanotube and thermal radiation. They realized that by enhancing the standards of the melting parameter, they were able to notice a greater velocity. Nowadays, finding new ways to improve thermal efficiency is an essential topic [14–24].

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Table 1. Details of current heat exchanger with MCTT.

Parameters	Length	Diameter	Thickness	Absorber plate thickness	Absorber plate width	MCTT length	MCTT thickness
Value (m)	2	0.025	0.001	0.0015	0.04	1.5	0.0005

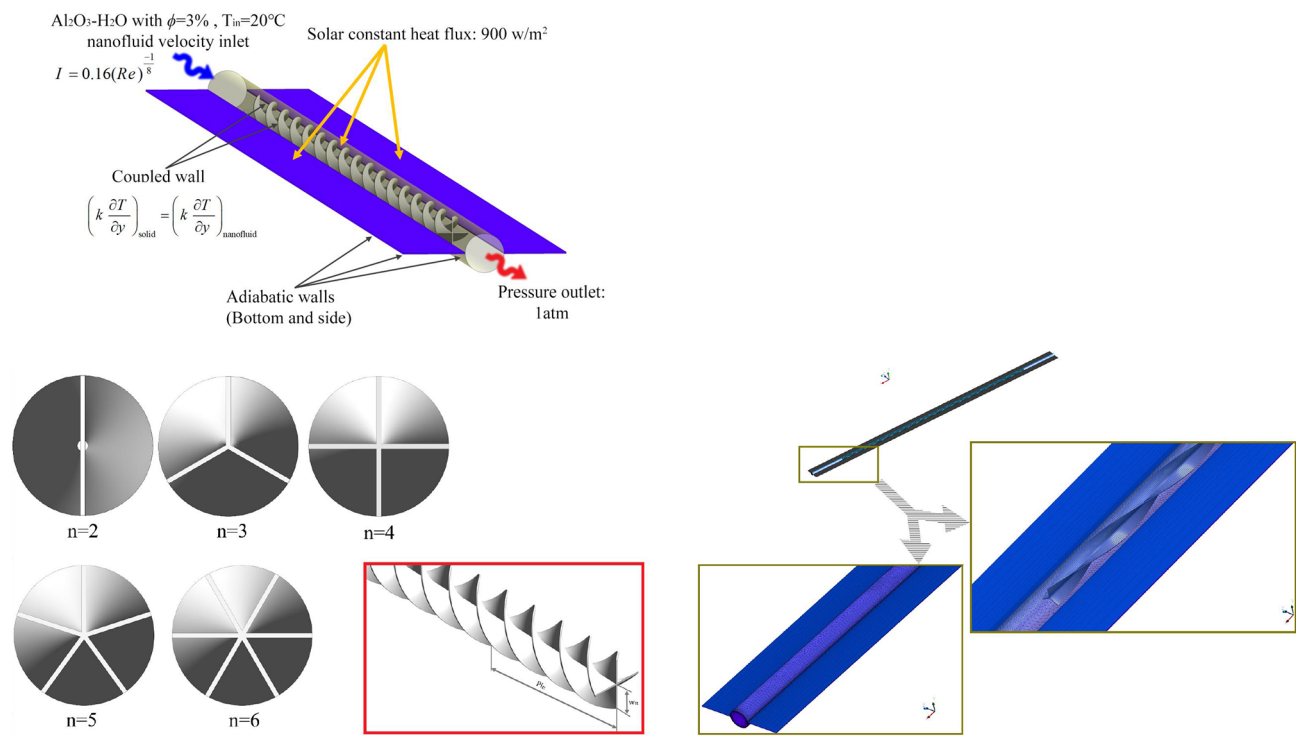


Fig. 1. Geometry of solar heat exchanger, boundary conditions and mesh.

Table 2. Thermo-physical properties of material.

Physical properties	Aluminum (solid)	Al ₂ O ₃	CuO	TiO ₂	H ₂ O	MWCNT
C_p (J/KgK)	871	765	531.8	686.2	4181.8	711
ρ (kg/m ³)	2719	3900	6320	4250	998.2	2100
k (W/mK)	202.4	30	76.5	8.9538	0.60	3000
μ (kg/ms)	—	—	—	—	0.001003	—

In this text, the MCTT has been employed to enhance the efficiency of heat exchangers. A pipe was connected to absorber plates and alumina was considered as the fluid. Our focus was on entropy generation. Simulations by means of the FVM have been reported to show the effect of the number of channel and revolution, the Reynolds number and the pipe diameters.

2 Current heat exchanger and mathematical model

In this article, a plain pipe with a length of 2 m and a diameter of 0.025 by using the MCTT has been numerically scrutinized. The pipe was connected to two absorber plates with a size of 0.04×1.5. The plate thickness equals 0.005 and plates are made up from Al. Table 1 illustrates the details of geometry. Figure 1 exhibits the current heat exchanger. Besides, we tried to write boundary conditions in that graph. The bottom side of plates and pipe is adiabatic while the upper side is under the constant heat flux with a value of 900 W. Such value is reasonable as an average during the day in Babol city. A mixture of water and Al₂O₃ with a concentration of 0.03, moves inside the pipe in the x direction. Table 2 shows material properties. Figure 1 depicts the schematic of a turbulator with various n . The temperature of the inlet nanofluid is 20 °C and turbulent intensity is assumed as $I = 0.16(Re)^{-1/8}$. For the outlet boundary, we

Table 3. Other details of the present code.

Pressure	second order
Momentum	first-order upwind
Turbulent kinetic energy	first-order upwind
Gradient	least squares cell based
Turbulent dissipation rate	first-order upwind
Pressure-velocity coupling	simple

Table 4. Variation of Y^+ when $n = 6$, $N = 7$, $D^* = 0.14$.

Y^+	Re
3.75	20000
3.19	16000
2.62	12000
2.03	8000
1.27	4000

considered the pressure outlet. The following equations can be taken into account for the current problem:

$$\frac{\partial(u_i)}{\partial x_i} = 0, \quad (1)$$

$$-\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \mu_{nf} \right) + \frac{\partial}{\partial x_j} \left(-\rho_{nf} \overline{u'_j u'_i} \right) = \frac{\partial}{\partial x_j} (\rho_{nf} u_i u_j), \quad (2)$$

$$\Gamma_t (= \mu_t / Pr_t), \quad \Gamma (= \mu_{nf} / Pr_{nf}), \quad \frac{\partial}{\partial x_i} \left((\Gamma + \Gamma_t) \frac{\partial T}{\partial x_i} \right) = \rho_{nf} \frac{\partial}{\partial x_i} (T u_i), \quad (3)$$

$$-\rho_{nf} \overline{u'_i u'_j} = \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \mu_t - \frac{2}{3} \rho_{nf} k \delta_{ij} - \frac{2}{3} \mu_t \frac{\partial u_k}{\partial x_k} \delta_{ij}, \quad (4)$$

$$\mu_t = \frac{1}{\varepsilon} k^2 C_\mu \rho_{nf}, \quad (5)$$

$$\frac{\partial}{\partial x_j} \left(\left(\frac{\mu_t}{\sigma_k} + \mu_{nf} \right) \frac{\partial k}{\partial x_j} \right) - \rho_{nf} \varepsilon + G_k = \frac{\partial}{\partial x_i} (u_i \rho_{nf} k), \quad G_k = -\rho_{nf} \overline{u'_j u'_i} \frac{\partial u_j}{\partial x_i}, \quad (6)$$

$$\frac{\partial}{\partial x_i} (u_i \rho_{nf} \varepsilon) = \frac{\varepsilon}{k} G_k C_{1\varepsilon} - \rho_{nf} \frac{\varepsilon^2}{k} C_{2\varepsilon} + \frac{\partial}{\partial x_j} \left(\left(\frac{\mu_t}{\sigma_\varepsilon} + \mu_{nf} \right) \frac{\partial \varepsilon}{\partial x_j} \right), \quad (7)$$

$$C_{1\varepsilon} = 1.42, \quad C_\mu = 0.0845, \quad C_{2\varepsilon} = 1.68, \quad Pr_t = 0.85, \quad \sigma_k = 1, \quad \sigma_\varepsilon = 1.3. \quad (8)$$

To predict the behavior of alumina nanofluid, the following formulations have been employed:

$$\rho_{nf} = \rho_f (1 - \phi) + \rho_s \phi, \quad (9)$$

$$\phi - 1 + \frac{(\rho C_p)_{nf}}{(\rho C_p)_f} = \frac{(\rho C_p)_s}{(\rho C_p)_f} \phi, \quad (10)$$

$$k_{nf} = k_f \frac{-2\phi k k + 2k_f + k_p}{\phi k k + k_p + 2k_f}, \quad k k = k_f - k_s, \quad (11)$$

$$\mu_{nf} = \mu_f / (1 - \phi)^{2.5}. \quad (12)$$

We employed the powerful turbulent model namely, “ K - ε , realizable”. To enhance the accuracy of the motioned model, the “enhanced wall treatment” option is also considered. Due to the assumption of incompressible flow, we simulated the problem with a pressure based solver. Other solution techniques and spatial discretization are summarized in table 3. In the selected model, Y^+ for all simulations should be lower than 5. As exhibited in table 4, Y^+ augments with increasing inlet velocity while the maximum value is 3.75 and satisfies the above limitation. To make a cautious choice of mesh size, all cases should be studied in view of mesh independence. To find a compromise between computational time and accuracy, the normal mesh can be selected for all the outcomes (table 5). A mesh sample is shown in fig. 1.

Table 5. Several mesh size for $D^* = 0.14$, $Re = 20000$, $N = 7$ and $n = 2$.

Mesh	Extra coarse	Coarse	Normal	Fine	Very fine
Number of elements	1 574 195	4 540 883	6 953 150	9 847 875	12 989 440
f	0.061847	0.058586	0.058030	0.057874	0.057164

To calculate the components of entropy generation and Be , Φ_s , the following equations can be employed:

$$\begin{aligned}
 S_{gen,total} &= S_{gen,th} + S_{gen,f} \\
 &= \frac{k_{nf}}{T^2} \left[\left(\frac{\partial T}{\partial x} \right)^2 + \left(\frac{\partial T}{\partial y} \right)^2 + \left(\frac{\partial T}{\partial z} \right)^2 \right] \\
 &\quad + \frac{\mu_{nf}}{T} \left\{ \left(\frac{\partial u_y}{\partial x} + \frac{\partial u_x}{\partial y} \right)^2 + \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x} \right)^2 + \left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y} \right)^2 + 2 \left[\left(\frac{\partial u_x}{\partial x} \right)^2 + \left(\frac{\partial u_y}{\partial y} \right)^2 + \left(\frac{\partial u_z}{\partial z} \right)^2 \right] \right\}, \quad (13)
 \end{aligned}$$

$$\phi_s = \frac{S_{gen,e}}{S_{gen,P}}, \quad (14)$$

$$Be = \frac{S_{gen,th}}{S_{gen,total}}. \quad (15)$$

3 Results and discussion

In this text, the impact of inserting the MCTT inside a pipe of solar heat exchanger has been evaluated. Entropy generation of alumina nanofluid has been described. We utilized such nanofluid due to previous researches about solar applications. Most articles employed aluminum oxide as an additive. The above side of pipe and absorbers experienced uniform solar heat flux.

To inform about details of the validation case, one can see fig. 2. The plain pipe with the same nanofluid which was selected by us was simulated [25]. The whole surrounding of wall was under the constant heat flux. The Reynolds number was considered to be 6020, which is indicated as a turbulent flow. Comparing the local convective coefficient demonstrates nice accuracy of our simulation. In addition, verification of the current code for the friction factor has been exhibited in fig. 3. We compared the outputs with those obtained via empirical formulas (Petukhov and Blasius) as defined below [26]:

$$f = (0.79 \ln Re - 1.64)^{-2}, \quad (16)$$

$$f = 0.318 Re^{-0.25}. \quad (17)$$

The working fluid was considered as the same as in our paper (Al_2O_3 -water nanofluid). This factor was calculated for various Re and reported in this figure. This graph depicts good agreement. Furthermore, to check the accuracy for pipe with twisted tape, we provided another validation in which results of previous experimental work [27] were reproduced. Geometric parameters are mentioned in fig. 4, we reported error percentage for various inlet velocities. As one can see in this graph, the maximum error was obtained lower than 4% which indicates the nice correctness of our simulation.

Components of entropy generation with their contours in various sections were presented in figs. 5 and 6. It is clear that the flow style in the presence of the MCTT is different from that of the plain pipe. As the number of channels augments, turbulent intensity augments which is related to more longitudinal vortexes. Thus, $S_{gen,th}$ reduces with a rise in n . Augmenting N results in higher flow mixing. So, the wall temperature reduces and this change makes $S_{gen,th}$ to decline. A greater diameter ratio makes the nanofluid revolve and leads to the suppression of the boundary layer. Higher swirl intensity can be gained with a rise in pumping power. Therefore, augmenting inlet velocity results in lower $S_{gen,th}$. If friction enhances, $S_{gen,f}$ augments. So, all active variables were directly proportional to $S_{gen,f}$.

Impacts of scrutinized parameters on turbulence intensity have been investigated in fig. 7. The main reason for utilizing a turbulator is its capacity to induce turbulence intensity next to the pipe wall. While increasing this parameter leads to higher flow resistance. As the convective flow enhances with a rise in pumping power, turbulent intensity enhances. Utilizing a greater number of channels can augment the intensity of turbulence near the hot surface. Vorticity augments with a rise in the diameter ratio, which indicates greater turbulent intensity. As the number of revolutions enhances, the thickness of the thermal boundary layer reduces and promotes turbulence intensity.

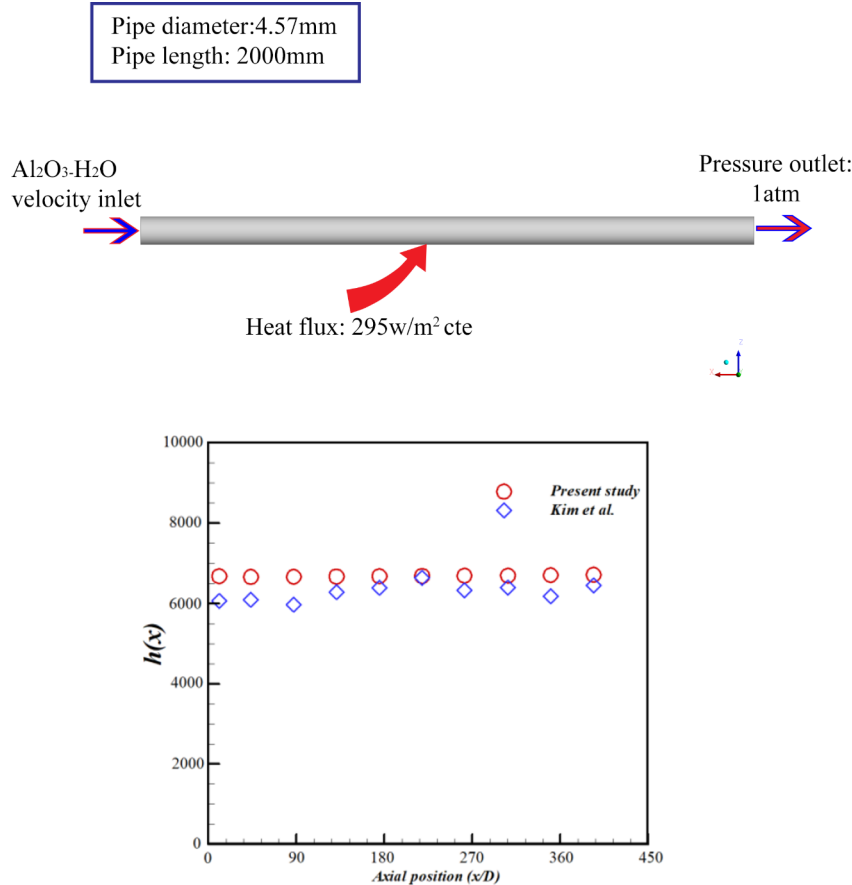


Fig. 2. Detail of the selected geometry for verification [25] at $Re = 6020$ and $\phi = 3\%$.

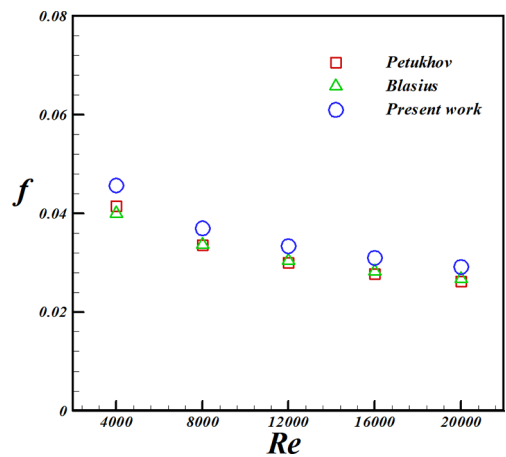


Fig. 3. Variation of the friction factor for a plain tube with Al₂O₃/water nanofluid.

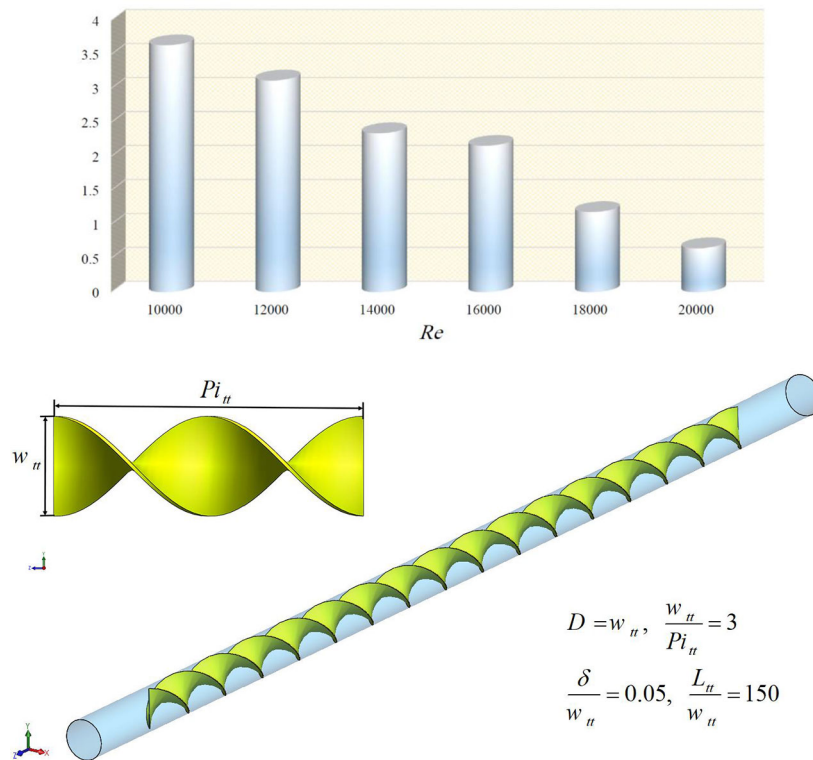


Fig. 4. Presentation of error percentage ($[(Nu - Nu_{exp})/Nu_{exp}] \times 100$) in comparison with ref. [27].

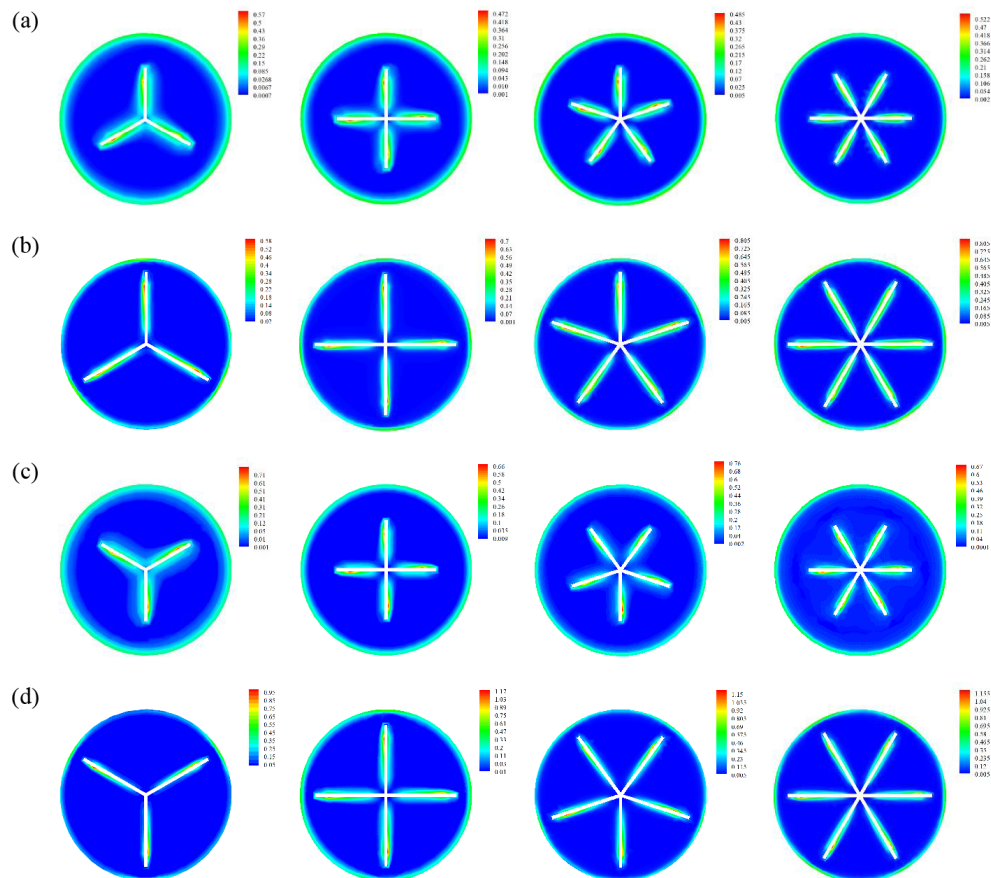


Fig. 5. Contour of $S_{gen,f}$ when $Re = 4000$ and $x/D = 40$: (a) $N = 4$, $D^* = 0.10$; (b) $N = 4$, $D^* = 0.14$; (c) $N = 7$, $D^* = 0.10$; (d) $N = 7$, $D^* = 0.14$.

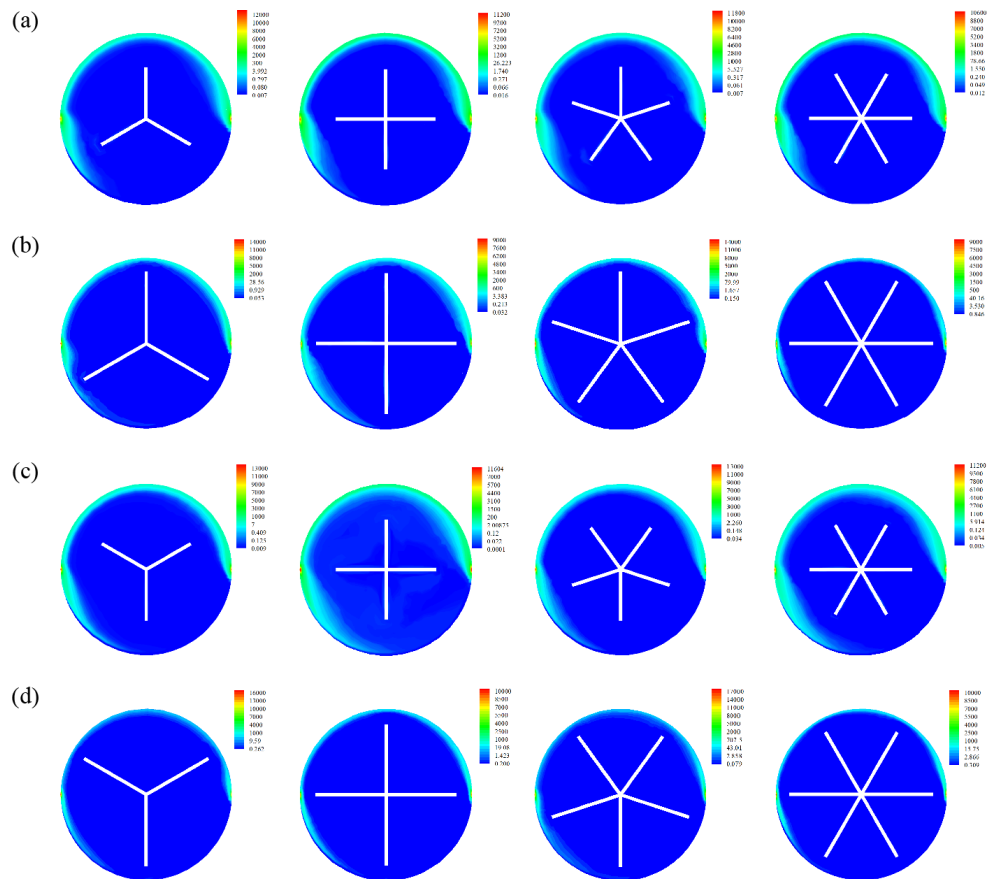


Fig. 6. Contour of $S_{gen,th}$ when $Re = 4000$ and $x/D = 40$: (a) $N = 4$, $D^* = 0.10$; (b) $N = 4$, $D^* = 0.14$; (c) $N = 7$, $D^* = 0.10$; (d) $N = 7$, $D^* = 0.14$.

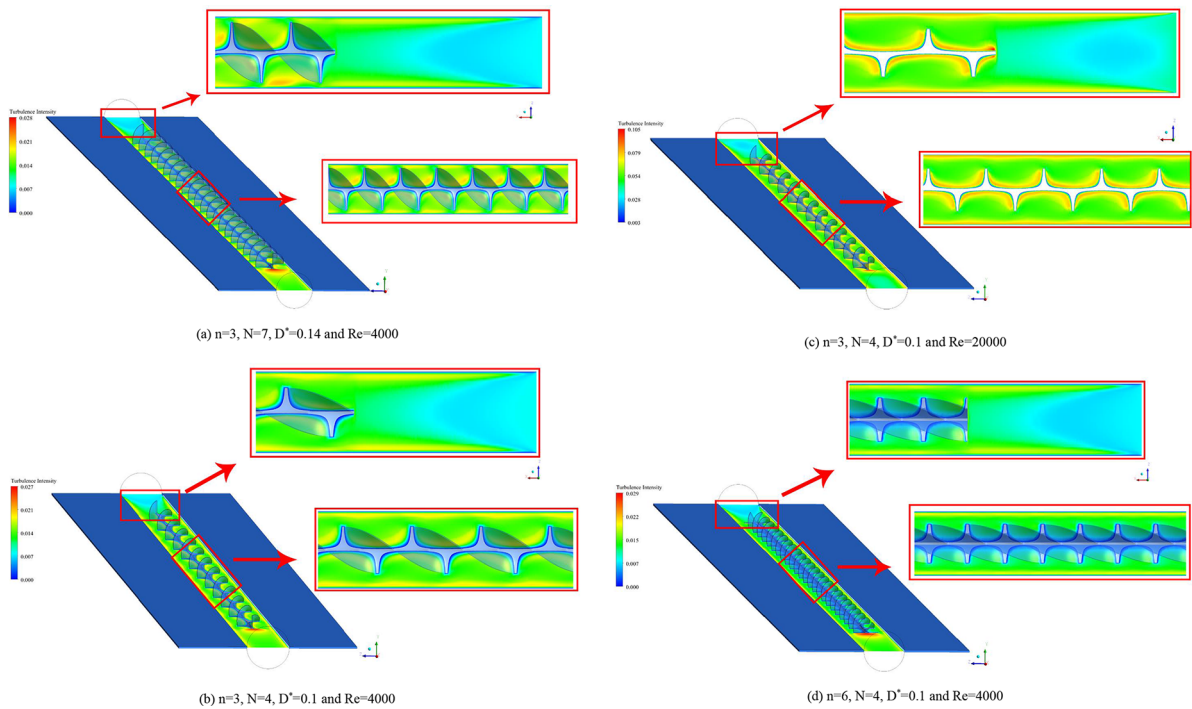
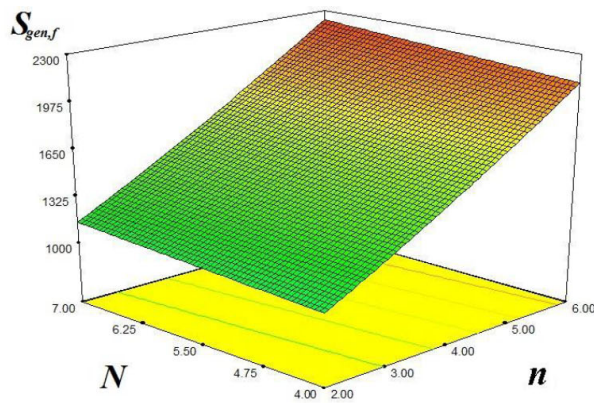
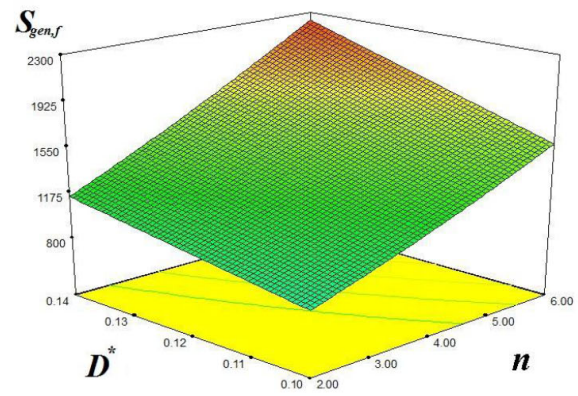
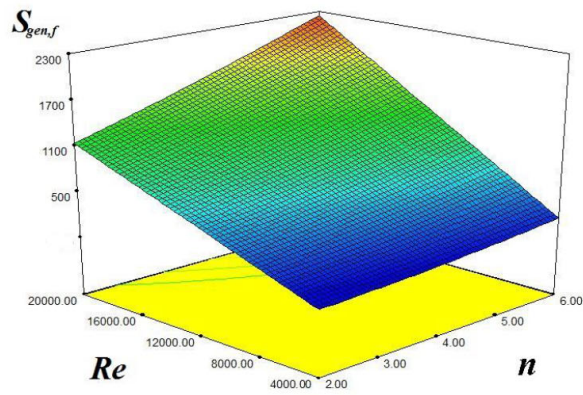
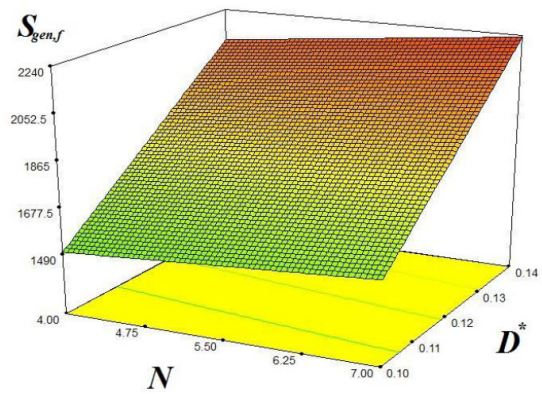
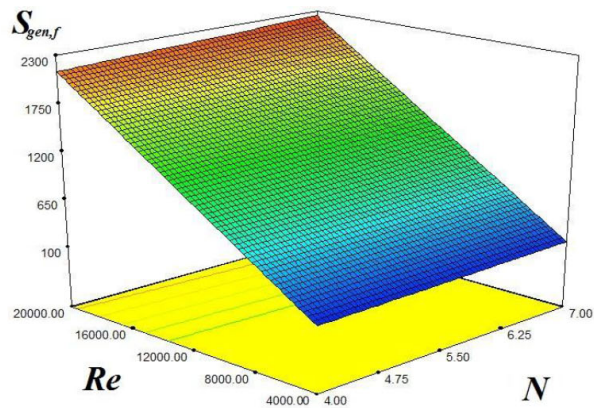
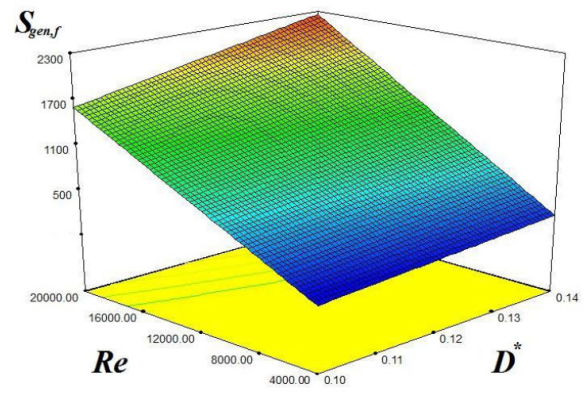


Fig. 7. Contour of turbulence intensity for various cases.

Re = 20000, $D^* = 0.14$  $N = 7, \text{Re} = 20000$  $N = 7, D^* = 0.14$  $n = 6, \text{Re} = 20000$  $n = 6, D^* = 0.14$  $n = 6, N = 7$ **Fig. 8.** Variation of $S_{gen,f}$ with respect to active parameters.

Influences of N , n , Re , D^* on average entropy, Be and Φ_s are illustrated in figs. 8, 9, 10 and 11. The attribution of more longitudinal eddies with augmentation of heat transfer results in this fact that $S_{gen,th}$ enhances with a rise in Re . Variation of Be can be explained with flow style changes. The augmenting number of revolutions leads to more radial velocity and more suppression of the boundary layer. So, share of $S_{gen,th}$ from $S_{gen,total}$ reduces and this means lower Be . More longitudinal eddies can be produced by increasing D^* . Thus, Φ_s declines with an increase in diameter. More nanofluid flow interactions can be seen in cases with a greater number of channels. So, Φ_s is inversely proportional to n .

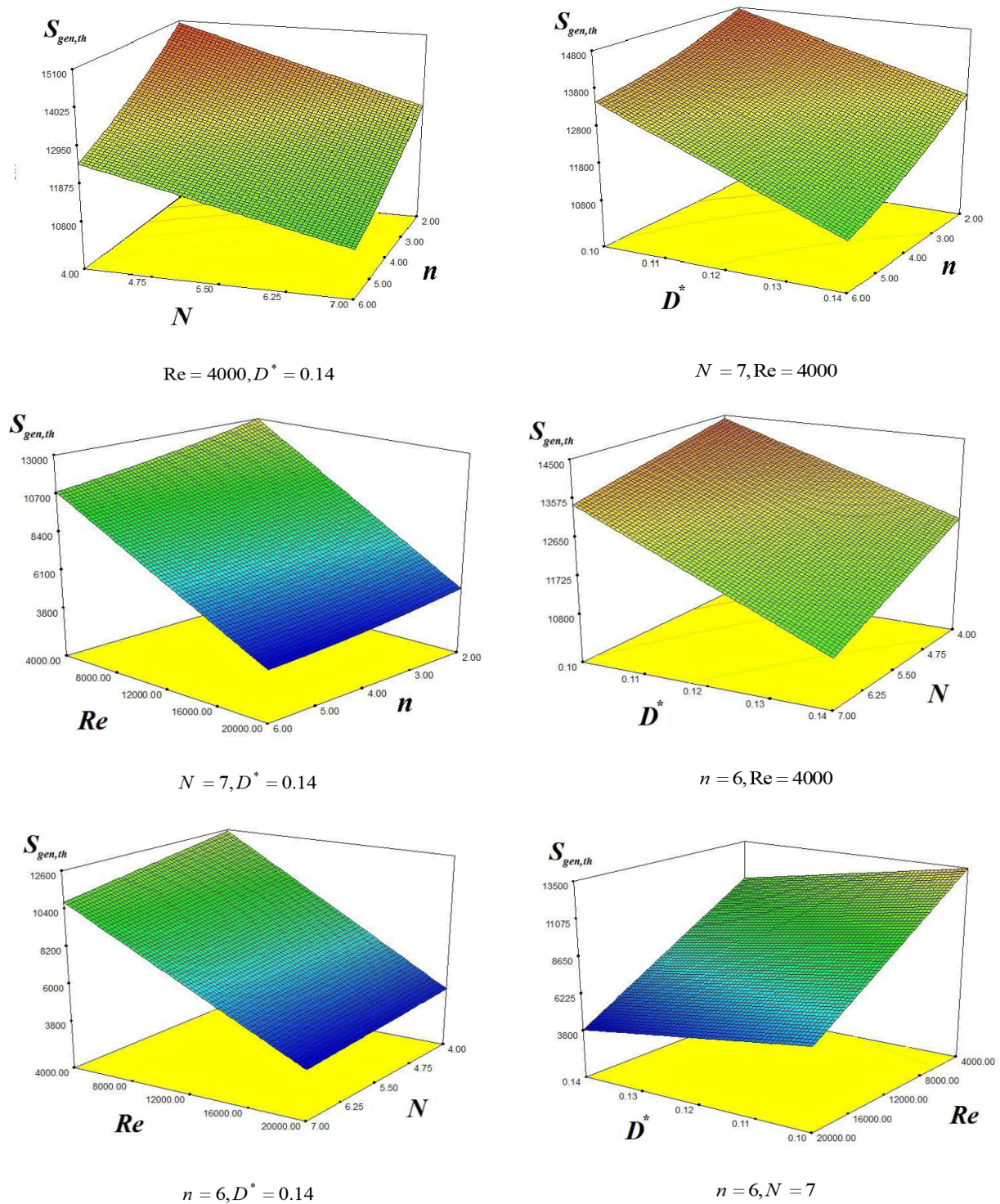
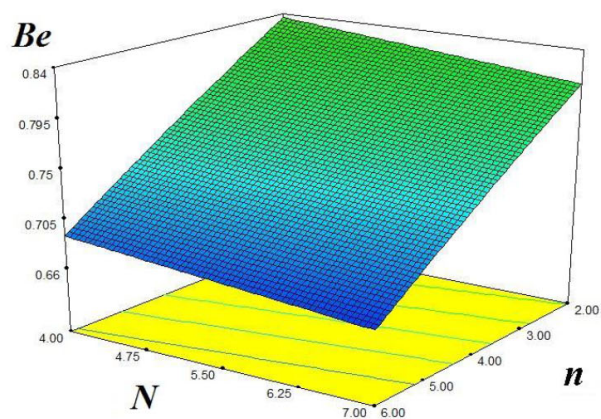
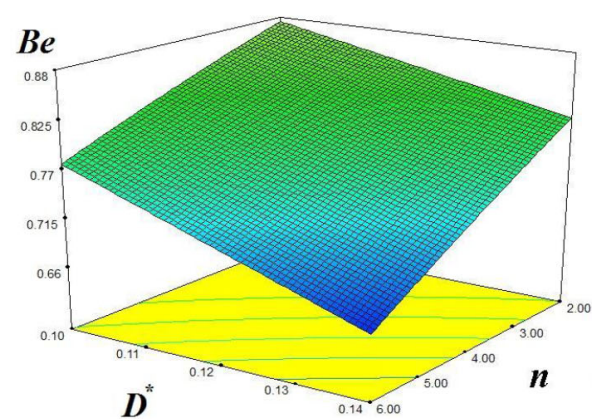


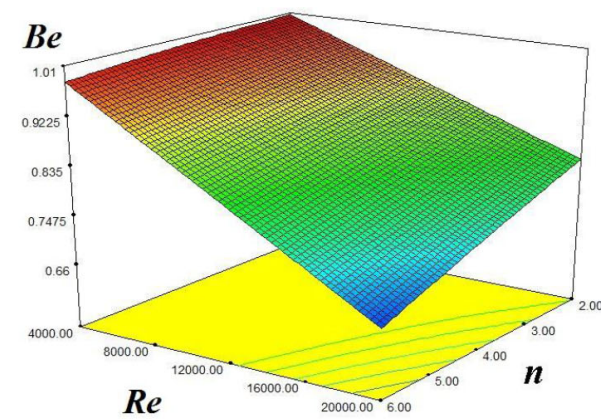
Fig. 9. Variation of $S_{gen,th}$ with respect to active parameters.



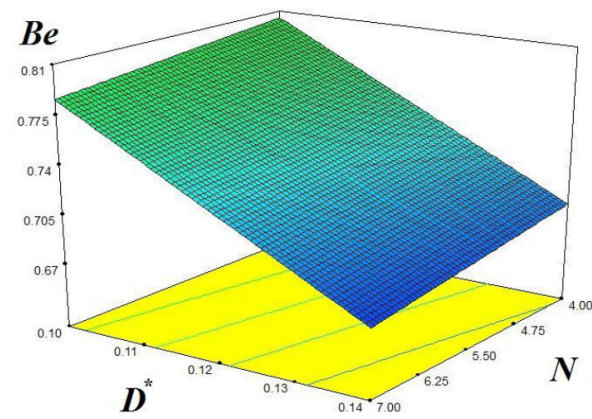
$Re = 20000, D^* = 0.14$



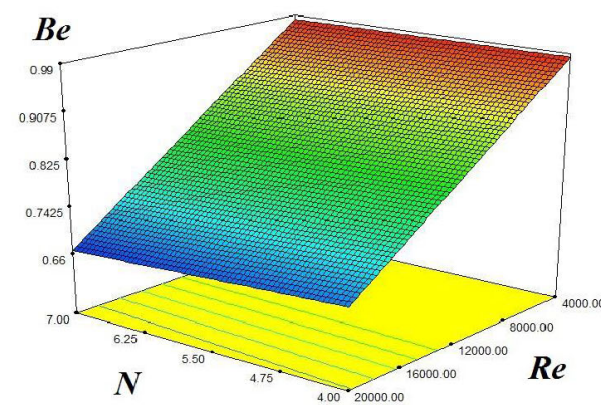
$N = 7, Re = 20000$



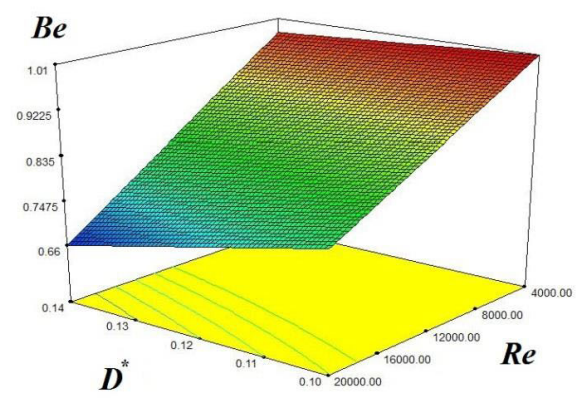
$N = 7, D^* = 0.14$



$n = 6, Re = 20000$



$n = 6, D^* = 0.14$



$n = 6, N = 7$

Fig. 10. Variation of Be with respect to active parameters.

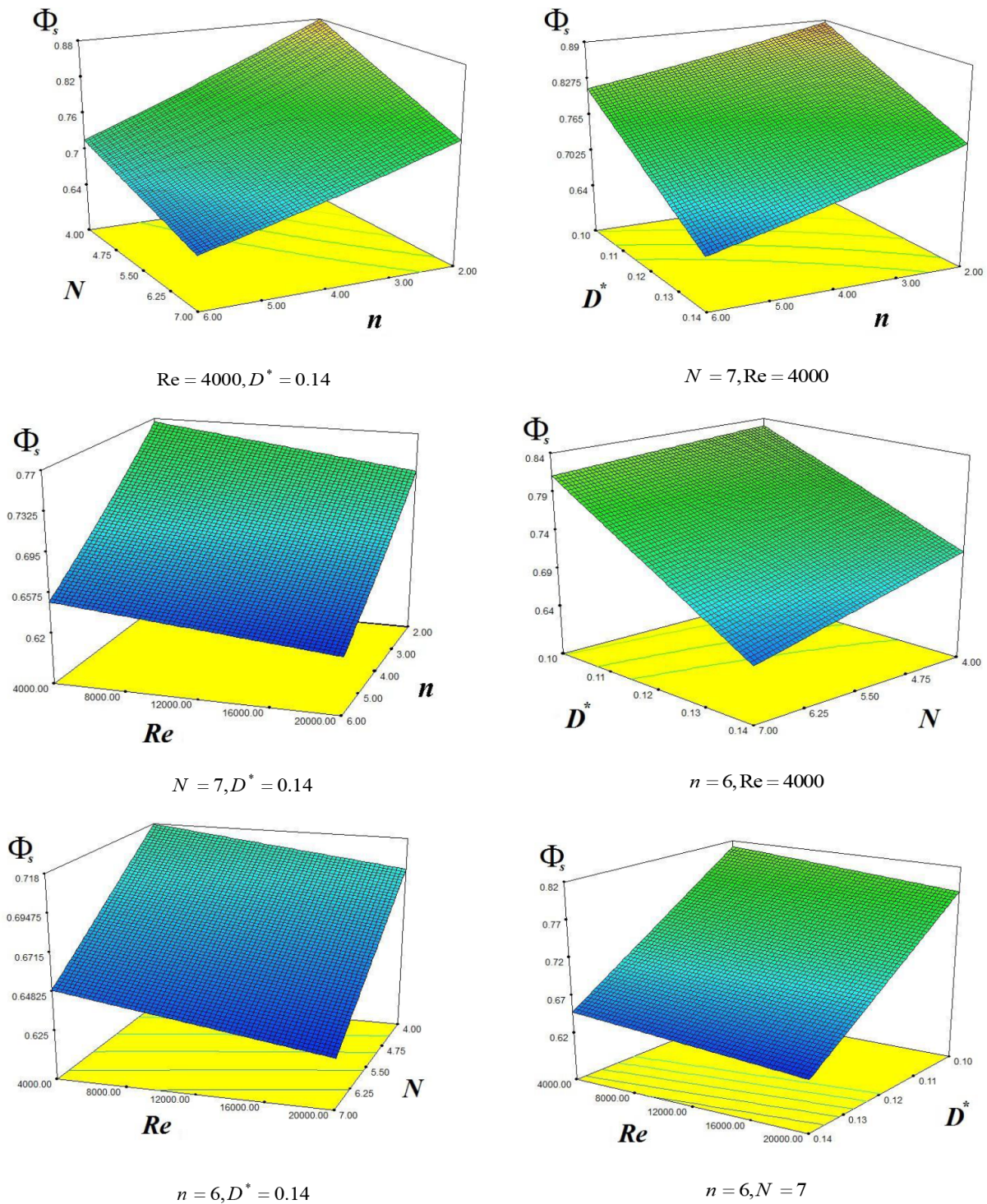


Fig. 11. Variation of Φ_s with respect to active parameters.

In this section we paid attention to the effect of nanoparticles' type. Figures 12 and 13 demonstrate impacts of nanoparticles' type on goal functions. The simulations reveal that the nanoparticles with higher k_{nf} generate higher turbulence intensity and a more secondary flow appears. Among the utilized nanoparticles, CuO has maximum density. So, maximum of $S_{gen,f}$ has been obtained for copper oxide. Adding nanoparticles leads to increase the friction resistance. Also, the base fluid has greater $S_{gen,th}$ in comparison with other cases. A minimum of $S_{gen,th}$ is reported for the MWCNT due to its highest k . Changing type of nanoparticles has no sensible impact on Be and its minimum values belong to the MWCNT. Φ_s is obviously changed with utilizing various materials. Its minimum and maximum values belong to the MWCNT and TiO_2 , respectively.

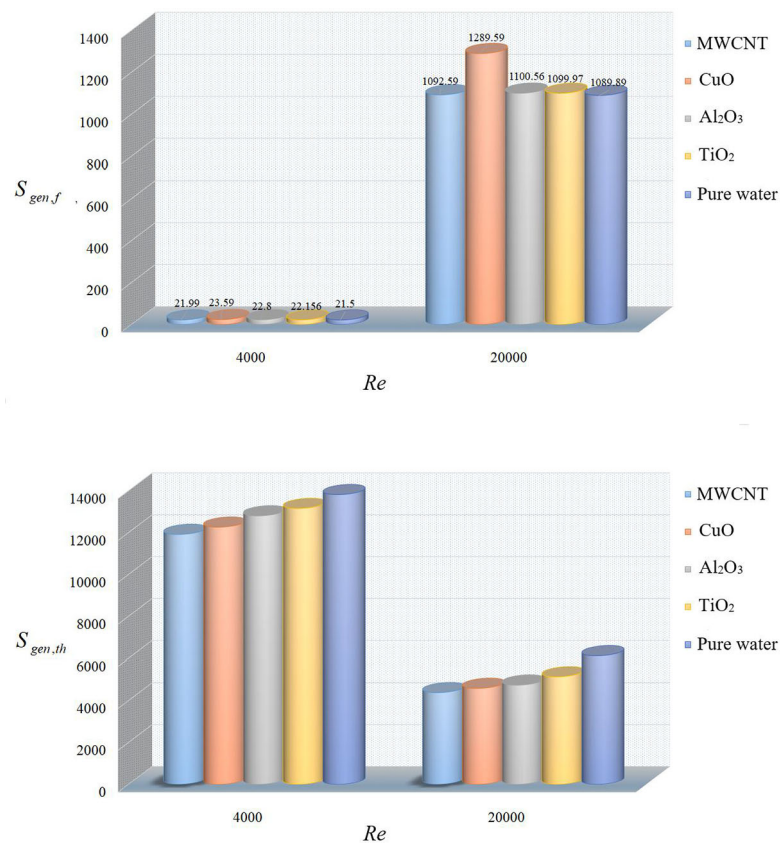


Fig. 12. Variation of entropy generation with changing type of nanoparticles when $n = 2$, $D^* = 0.14$, $N = 7$.

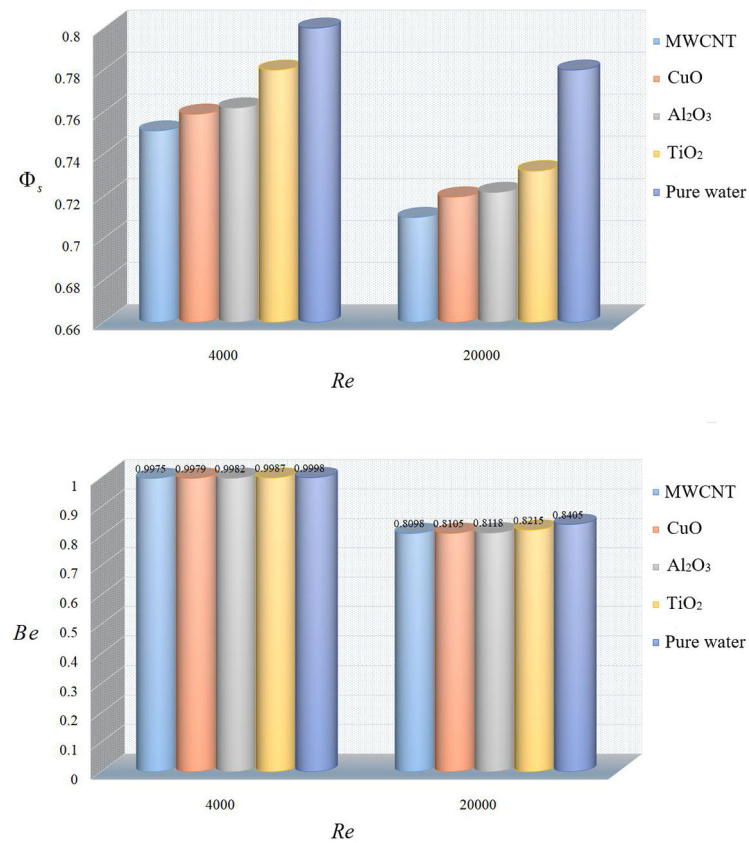


Fig. 13. Variation of Φ_s and Be with changing type of nanoparticles when $n = 2$, $D^* = 0.14$, $N = 7$.

4 Conclusion

In this report, the MCTT was imposed into the solar heat exchanger. Aluminum oxide was mixed with water and acted as the working fluid. Solar radiation was employed as the constant heat flux which affects the above side of the absorber and pipe. As the convective flow enhances with a rise in pumping power, turbulent intensity enhances. Scrutinized parameters were inversely proportional to the variation of the Bejan number. Greater radial velocity was produced with selecting greater N . Minimum of $S_{gen,th}$ was reported for the MWCNT due to its highest k . $S_{gen,th}$ declines with a rise in pumping power due to the reduction in surface temperature.

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Nomenclature

L	length of pipe	Greek symbols	
FVM	finite volume method	ρ	density
G_k	rate of turbulent kinetic energy generation	ϕ	concentration of nanoparticles
q''	heat flux	μ	dynamic viscosity
Nu	Nusselt number	Φ_s	augmented entropy generation
Re	Reynolds number	Γ_t	turbulent thermal diffusivity
N	revolution ($= L_{tt}/Pi_{tt}$)	ε	dissipation rate
Pr	Prandtl number	α	thermal diffusivity
Pi	pitch length	Subscripts	
p	pressure	s	solid
T	fluid temperature	nf	working fluid
f	Darcy friction factor	e	enhanced tube
MCTT	multi-channel twisted tape	tt	twisted tape
K	turbulent kinetic energy	s	particles
D^*	diameter ratio ($= 20w_{tt}/L_{tt}$)	o	operating
Be	Bejan number	f	water

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